

Beyond Retrieval: Analytic Memory for Multimodal Agents

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Abstract

Long-term multimodal memory must support not only retrieving relevant information but also computing over observations accumulated across interactions. Existing systems largely emphasize *retrieval memory*, organizing interaction histories through summaries and indexes to return query-relevant information at multiple granularities, from high-level abstractions to underlying records. In this paper, we formulate *analytic memory* as a complementary abstraction that organizes recurring multimodal observations into queryable structures supporting filtering, aggregation, ranking, and temporal comparison. We present ADAMM, a framework that jointly supports retrieval and analytic memory. Rather than relying on application-defined schemas, ADAMM extracts provenance-linked attribute-value observations from dialogue, images, and contextual metadata, discovers recurring field structures, and materializes them for analytical access. At inference time, a memory-aware planner decomposes queries into retrieval and analytic operations and routes each operation to the appropriate tools. Experiments on two long-term multimodal memory benchmarks, MemEye and MemGallery, show that ADAMM improves performance by up to 11.3% and 7.3%, respectively.

1 Introduction

As Large Language Model (LLM) agents interact with users over increasingly long horizons, they accumulate extensive multimodal histories – including dialogues, screenshots, images, documents, and contextual observations – that quickly exceed the finite context windows of their backbone models (He et al., 2024; Koh et al., 2024; Xie et al., 2024; Wang et al., 2024). Multimodal memory systems, which retain and organize such histories to support future reasoning (Wang and Chen, 2025; Feng et al., 2026), have therefore become essential

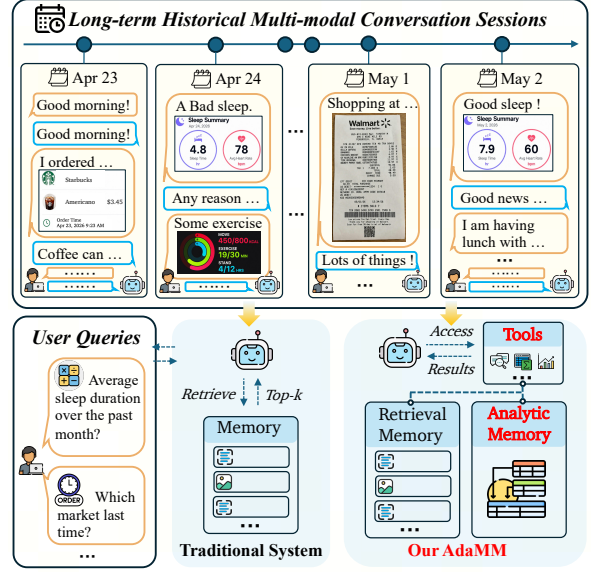


Figure 1: Long-term multimodal interaction histories contain observations distributed across sessions, modalities, and time. Answering user queries may require either retrieving specific past events or performing analytical operations, such as temporal selection and aggregation, over multiple observations.

and are widely adopted in applications such as personal assistance (Li et al., 2024; Feng et al., 2026), embodied interaction (Driess et al., 2023; Wang et al., 2023), and tool-augmented workflows (Yao et al., 2022; Schick et al., 2023).

Recent multimodal memory systems have explored a broad range of designs, including textual abstraction (Feng et al., 2026), specialized memory types (Nguyen et al., 2026), hybrid stores (Wang and Chen, 2025), and cross-modal retrieval (Lu et al., 2026). Most existing systems follows a retrieve-then-answer paradigm, conditioning an LLM on a bounded set of relevant memories, which we denote as retrieval memory. Such systems are effective at selecting relevant memories from long interaction histories. However, long interaction histories also accumulate recurring observa-

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tions that collectively form an append-only log. Analytical questions over such histories require complete, correctly scoped records and operations such as filtering, aggregation, ranking, and temporal selection. Relevance-based approach adopted by retrieval memory systems therefore creates a coverage–context trade-off: a small retrieval set may omit required observations, whereas a larger retrieval set consumes the limited context budget and introduces redundant or distracting evidence. For example, as illustrated in Figure 1, computing a user’s average sleep duration over the past month requires collecting all relevant measurements across multiple screenshots before aggregation; incomplete retrieval may therefore produce a biased estimate. We call this gap between relevance-based retrieval and analytic the *retrieval–analysis mismatch*.

To bridge this gap, our key insight is that a memory system should couple two complementary subsystems: *retrieval memory*, which identifies and supplies relevant historical records as context, and *analytic memory*, which proactively organizes interaction histories into reusable and queryable structures and adaptively executes operations over them. Together, they support both flexible semantic recall and structured analysis, rather than requiring the LLM to repeatedly reconstruct the needed organization from retrieved records. Realizing analytic memory, however, presents two key challenges. First, interaction histories do not come with an explicit organizing schema, making it unclear what information should be preserved and how they should be structured to support future queries. Second, user queries rarely specify memory access strategies, the system therefore must reconcile what each query requires with what the evolving memory can currently support. Even questions over the same underlying records may require fundamentally different operations (e.g., aggregation, ordering, filtering), posing challenges to the design of adaptive execution plans.

To address these challenges, we introduce ADAMM, an **Adaptive Multi-view Memory** framework that couples data-driven structure induction with memory-aware query planning. It maintains two complementary subsystems: *retrieval memory* for flexible semantic access and *analytic memory* for executable analysis over recurring observations. ADAMM extracts provenance-linked attribute–value pairs from multimodal interactions (e.g., *{Sleep Time: 5.5 hrs}*), induces recurrent

schemas, and materializes them as structured tables, which is accessible via tools, while preserving unstructured evidence in a hierarchical semantic graph. At query time, a planner adaptively composes different analytic and retrieval tools based on the query and the memory structures currently available. This design enables grounded semantic recall and structured analysis over long-term interaction histories.

Our contributions are summarized as follows:

- We identify a *retrieval–analysis mismatch* in long-term multimodal memory: retrieval-oriented interfaces support returning query-relevant information at multiple granularities, but cannot reliably answer queries requiring complete-range filtering, aggregation, ranking, or temporal selection over append-only interaction histories.
- We propose ADAMM, a multimodal memory framework that complements retrieval memory with schema-induced analytic memory. ADAMM discovers recurrent structures from multimodal observations, materializes them as queryable tables, and uses a memory-aware planner to select or compose semantic retrieval with native analytical operations.
- We evaluate ADAMM on two multimodal long-term memory benchmarks, MemEye and MemGallery, where ADAMM consistently outperforms strong memory baselines, and improves accuracy by up to 11.3% and 7.3%, respectively.

2 Related Work

Agent memory. Long-term agent memory preserves information across interactions, allowing agents to reuse preferences, observations, decisions, and task states beyond a finite context window (Hatalis et al., 2023; Li et al., 2024; Zhang et al., 2025). Existing systems explore personalized stores and tiered memory management (Zhong et al., 2024; Packer et al., 2023), scalable or agentic memory construction (Chhikara et al., 2025; Xu et al., 2025), and temporal or relational organization (Rasmussen et al.; Huang et al., 2025). Recent schema-grounded memory moves beyond semantic recall by transforming textual interactions into validated, queryable records, but relies on application-defined schemas (Petrov et al., 2026). Despite these advances, existing methods remain largely text-centric, leaving underexplored how to

induce reusable analytical structures from heterogeneous multimodal histories and adaptively execute retrieval and analytical operations over them.

Multimodal agent memory. Multimodal agent memory extends text-centric memory to histories containing heterogeneous visual observations. Existing approaches broadly follow two directions: MIRIX and M2A translate visual content into captions, summaries, or semantic abstractions for text-based retrieval (Wang and Chen, 2025; Feng et al., 2026), whereas MMA, VisualMem, and OmniSimpleMem retain native visual evidence and retrieve multimodal or image-backed memories (Lu et al., 2026; Nguyen et al., 2026; Liu et al., 2026). Although these methods improve visual evidence preservation, they still primarily organize histories as semantic memories. MemEye further reveals their difficulty in capturing fine-grained details and evolving visual states (Guo et al., 2026). In contrast, ADAMM adaptively organizes multimodal histories into complementary analytical and semantic memories, enabling both precise analysis and context-aware retrieval for complex queries.

3 Method

3.1 Problem Formulation

Let $\mathcal{H} = \{S_i\}_{i=1}^M$ denote a multimodal interaction history, where each session $S_i = \{R_{i,j}\}_{j=1}^{n_i}$ consists of multiple interaction rounds. Each round $R_{i,j} = (d_{i,j}, \mathcal{V}_{i,j}, \tau_{i,j})$ comprises the user-agent dialogue $d_{i,j}$, visual observations $\mathcal{V}_{i,j}$, and temporal information $\tau_{i,j}$. A multimodal agent memory system transforms the interaction history into memory representations, $\mathcal{M} = F_{\text{build}}(\mathcal{H})$, and aims to generate an accurate answer $\hat{y} = F_{\text{answer}}(\mathcal{M}, q)$ for a user query q based on $\mathcal{M}$.

3.2 Overview

Figure 2 presents the overview of the proposed ADAMM framework. Given a long-term multimodal interaction history, ADAMM constructs two complementary memories: analytic memory $\mathcal{M}^{\text{ana}}$ induces recurring patterns as adaptive schemas for analytical queries while retrieval memory $\mathcal{M}^{\text{ret}}$ hierarchically organizes interactions according to their semantics to enable flexible retrieval. At query time, ADAMM jointly reasons over the query and the memories instantiated from the current history, then constructs an execution plan by adaptively selecting and composing access tools to gather sufficient evidence for grounded answer generation.

3.3 Analytic Memory Construction

Analytic memory aims to uncover and organize recurring patterns across fragmented observations in multimodal interaction histories for analytical operations. The key challenge is that the meaningful attributes, which is in the form of key-value pair, and their co-occurrence patterns are unknown in advance, and each round reveals only a sparse fragment of the latent structure. To address this challenge, ADAMM treats $\mathcal{H}$ as a chronologically ordered interaction rounds $\{R_t\}_{t=1}^N$, performs *Attribute Extraction* to adaptively identify attribute evidence within each round, then conducts *Schema Induction* to discover stable structures from recurring cross-round patterns, and finally applies *Memory Materialization* to transform the induced schemas into executable analytic memory.

3.3.1 Attributes Extraction

Recovering recurring structure must begin from the evidence available within individual rounds, yet the attributes that will form a useful schema are unknown at extraction time. Record fragment extraction therefore aims to preserve explicit analytic observations from each round without imposing a predefined and unobserved pattern.

Given R_t , ADAMM jointly examines its dialogue and visual content to identify every grounded attribute-value correspondence:

$$\mathcal{O}_t = \text{Extractor}(R_t) = \{(a_{t\ell}, x_{t\ell}, p_{t\ell})\}_{\ell=1}^{m_t}, \quad (1)$$

where $a_{t\ell}$ and $x_{t\ell}$ are the observed attribute and its source-faithful value, and $p_{t\ell}$ points to the supporting dialogue span or image. Extractor is an LLM-based extractor that targets at identifying all possible pairs from each round. The resulting collection $\{\mathcal{O}_t\}_{t=1}^N$ captures these observations and their within-interaction co-occurrence patterns for subsequent schema induction.

3.3.2 Schema Induction

Given the extracted record fragments, it remains challenging to induce reusable schemas without assuming a predefined schema. New observations may reveal either a previously unseen structure or evolution to an existing one. To address this challenge, ADAMM first mine candidate patterns and then induces schemas through two processes: *Schema Discovery* creates new schemas, while *Schema Evolution* extends existing ones with consistently co-occurring attributes.

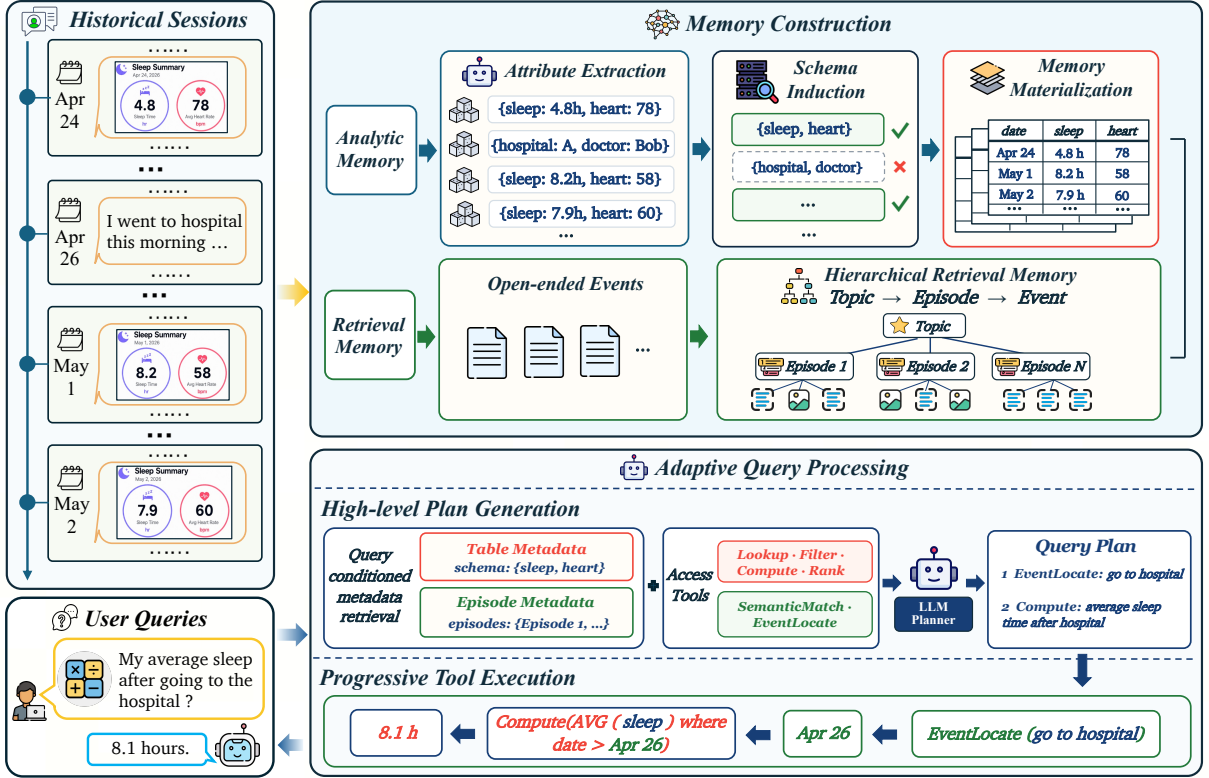


Figure 2: Overview of ADAMM. It complements recall-oriented retrieval memory with schema-induced analytic memory over recurrent multimodal observations. At query time, an operation planner jointly considers the query and the instantiated memories to select designed access tools for answer generation.

Candidate Pattern Mining. For each extracted record fragment $\mathcal{O}_t$, let

$$\mathcal{A}_t = \{a_{tl} : (a_{tl}, x_{tl}, p_{tl}) \in \mathcal{O}_t\} \quad (2)$$

denote the set of attributes contained in $\mathcal{O}_t$. A reliable candidate pattern $\mathcal{C}$ is a set of attributes that consistently recur sufficient interactions together. Following the Apriori framework for frequent-itemset mining (Agrawal and Srikant, 1994), we use *support* to quantify information recurrence. Specifically, the *support* of a candidate pattern $\mathcal{C}$ after processing round t is

$$\text{supp}_t(\mathcal{C}) = \sum_{s=1}^t \mathbb{I}[\mathcal{C} \subseteq \mathcal{A}_s], \quad (3)$$

ADAMM retains $\mathcal{C}$ as a candidate pattern if

$$\text{supp}_t(\mathcal{C}) \geq \theta_s, \quad (4)$$

which ensures that $\mathcal{C}$ recurs across sufficient interaction rounds to yield a populated schema.

Schema Discovery. For each recurrent candidate $\mathcal{C}$, ADAMM first compares it with the active schemas before round t . If $\mathcal{C}$ exhibits an attribute

structure distinct from all existing schemas, it is evaluated for reliable new-schema discovery. Following a variant of Apriori algorithm (Omiecinski, 2003), we use *all-confidence* to measure within-pattern co-occurrence consistency, defined as

$$\text{all-conf}_t(\mathcal{C}) = \frac{\text{supp}_t(\mathcal{C})}{\max_{a \in \mathcal{C}} \text{supp}_t(\{a\})}. \quad (5)$$

A candidate is admitted as a new schema if

$$\text{all-conf}_t(\mathcal{C}) \geq \theta_a. \quad (6)$$

This constraint requires the complete candidate to occur reliably whenever any of its constituent attributes occurs, thereby preventing a ubiquitous attributes from being merged with most patterns. To avoid admitting a qualified candidate together with its qualified subsets, ADAMM retains only inclusion-maximal novel candidates.

Schema Evolution. For an existing schema $\mathcal{A}$, if $\mathcal{C} \subseteq \mathcal{A}$, then $\mathcal{C}$ is already subsumed by it. If $\mathcal{C}$ contains $\mathcal{A}$, ADAMM treats $\mathcal{C}$ as a potential extension of $\mathcal{A}$. Let $\Delta = \mathcal{C} \setminus \mathcal{A}$ is the newly added attributes. We use *extension confidence* to measure

how consistently Δ accompany occurrences of $\mathcal{A}$, defined as

$$\text{ext-conf}_t(\mathcal{C} \mid \mathcal{A}) = \frac{\text{supp}_t(\mathcal{C})}{\text{supp}_t(\mathcal{A})} \quad (7)$$

When $\text{ext-conf}_t(\mathcal{C} \mid \mathcal{A}) \geq \theta_e$, the candidate provides sufficient evidence that Δ have become stable components of $\mathcal{A}$, and we update the schema $\mathcal{A}$ to $\mathcal{C}$. Otherwise, $\mathcal{A}$ remains unchanged.

3.3.3 Memory Materialization

Considering that each schema specifies a stable set of attributes and each matched interaction provides their co-occurring values, the induced patterns naturally align with relational tables. ADAMM therefore materialize each schema $\mathcal{A}$ into a table T_A , using its attributes as columns and the matched interactions as rows.

Specifically, for a newly discovered schema $\mathcal{A}$, its attributes first define the data columns. Given these attributes and their sampled values from the corresponding extracted observations $\{\mathcal{O}_t\}$, a language model generates the table metadata, including its name and description, and infers the type of each column. We then populate the table at the interaction level. For each associated round R_t , every extracted tuple $(a_{t\ell}, x_{t\ell}, p_{t\ell}) \in \mathcal{O}_t$ whose attribute belongs to $\mathcal{A}$ is mapped to the corresponding column, and values extracted from the same round jointly form one row. Two auxiliary columns, order and time, record the position and occurrence time of the source interaction, respectively. Any value not observed in that round is left empty. When a schema evolves from $\mathcal{A}$ to $\mathcal{A}' = \mathcal{A} \cup \Delta$, the attributes in Δ are appended as new columns to the existing table T_A . ADAMM then updates the table information and rematerializes the rows from their associated observations following the same procedure. The resulting analytic memory is represented as the collection of tables $\mathcal{M}^{\text{ana}} = \{T\}$.

3.4 Retrieval Memory Construction

Beyond structured analytic queries, many user requests can be addressed through direct semantic matching against past interactions. Retrieval memory is designed for such queries by preserving open-ended events, relations, and visual details that may not exhibit recurring schemas. It complements analytic memory by enabling flexible semantic retrieval and context-aware reasoning over interaction histories.

Following prior work on hierarchical memory organization (Yue et al., 2026), we construct retrieval memory $\mathcal{M}^{\text{ret}}$ into three levels: topic $\rightarrow$ episode $\rightarrow$ event. Events preserve fine-grained evidence extracted from individual interactions, episodes group temporally adjacent and semantically coherent events, and topics aggregate related episodes into broader semantic contexts. Each level maintains a concise description, multimodal representation, temporal information, and provenance. This hierarchy allows the planner to first identify relevant topics or episodes and then locate supporting events, preserving both high-level context and fine-grained evidence for complex queries.

3.5 Adaptive Query Processing

Different queries place different demands on memory: some require semantic retrieval over open-ended events, whereas others require structured computation over analytic memory. No single retrieval primitive can adequately support this full range of needs, calling for an adaptive mechanism that selects and composes operations according to query intent and the structures available in memory. ADAMM therefore abstracts the native capabilities of analytic and semantic memory as operation-specific tools, and employs a joint query-memory planner to generate an executable procedure for answering each query.

3.5.1 Memory Access Tools

Analytic and semantic memory differ in their underlying structures and consequently provide distinct native capabilities. To expose these capabilities at query time, ADAMM equips each memory with structure-specific tools.

Specifically, each tool is represented as

$$\xi_i = (d_i, \Theta_i, \Omega_i), \quad (8)$$

where d_i textually specify its tool function, while Θ_i and Ω_k define its typed arguments and return schema. Table 1 summarizes the access tools designed for analytic and semantic memory. For example, for the analytic tool COMPUTE, d_i specifies a deterministic computation over an induced table, Θ_i includes the target table, computation operator, target column, and optional constraints, and Ω_i returns the computed result and matched records. For the semantic tool SEMANTICMATCH, d_k specifies relevance-based memory retrieval, Θ_i includes a semantic query content and retrieval budget, and

Table 1: Access tools for analytic and retrieval memory.

Memory	Tools	Capability
Analytic	LOOKUP, FILTER, COMPUTE, RANK	Exact record access and deterministic computation.
retrieval	SEMANTICMATCH, EVENTLOCATE	Relevance-based retrieval and event localization.

Ω_i returns the ranked memory units with their relevance scores.

3.5.2 Memory-Aware Joint Query Planning

The access tools specify how each memory can be queried. However, users describe desired outcomes rather than access procedures, and a feasible operation also depends on what the current memory contains. To bridge this gap, ADAMM first constructs a query-conditioned planning context that exposes relevant memory structure, and then employs an LLM-based planner to jointly reason over the query and this context, composing an executable procedure for retrieving and analyzing the required information. The resulting evidence is subsequently provided to the answer model for response generation.

Planning Context Construction. Valid query planning requires knowledge of the current memory structures and their supported operations. Exposing the complete memory contents, however, introduces irrelevant information and expands the planning space. ADAMM therefore abstracts analytic and retrieval memory into compact metadata, retrieves query-relevant entries, and associates them with their available tools to form the effective planning context.

Specifically, we treat each analytic table $T \in \mathcal{M}^{\text{ana}}$ and retrieval episode $P \in \mathcal{M}^{\text{ret}}$ as an individual planning candidate represented by textual metadata μ_T and μ_P , respectively. For each table, μ_T summarizes its name, description, columns, and sampled values, while each episode directly uses its description as μ_P . Together, these candidates form the metadata search space $\mathcal{U}$. Given a query q , ADAMM ranks each $\mu \in \mathcal{U}$ using

$$h_\lambda(q, \mu) = \lambda \cos(E(q), E(\mu)) + (1 - \lambda) \text{TokSim}(q, \mu), \quad (9)$$

where $E(\cdot)$ is a semantic encoder and TokSim measures token-level similarity. After retaining the

top-ranked μ , ADAMM augments their metadata with designed access tool descriptions, forming the planning context $\mathcal{D}_q$ to expose relevant memory candidates and available operations to the planner.

Progressive Tool-Execution Planning. Complex queries often require multiple dependent operations, with the arguments of later tool calls determined by earlier outputs. Instantiating all tool calls upfront is therefore unreliable. To address this challenge, ADAMM separates high-level planning from progressive tool instantiation.

High-level plan generation. Conditioned on the query q and planning context $\mathcal{D}_q$, ADAMM employs an LLM-based planner to generate an abstract plan specifying the information goals and their execution order, without binding them to concrete tools or arguments:

$$\bar{\pi}(q, \mathcal{D}_q) = \text{Planner}(q, \mathcal{D}_q) = [(g_\ell, \xi_\ell)]_{\ell=1}^L, \quad (10)$$

where g_ℓ is a textual description of the information required at step ℓ , and ξ_ℓ specifies the access tool designated to obtain it.

Progressive tool instantiation. Guided by the above high-level plan, ADAMM instantiates tool calls sequentially. At step ℓ , it conditions on the preceding results $\mathcal{Z}_{<\ell} = \{z_1, \dots, z_{\ell-1}\}$:

$$\begin{aligned} \Theta_\ell &= \text{Planner}(q, g_\ell, \xi_\ell, \mathcal{D}_q, \mathcal{Z}_{<\ell}), \\ z_\ell &= \text{Invoke}(\xi_\ell, \Theta_\ell), \end{aligned} \quad (11)$$

where ξ_ℓ and Θ_ℓ denote the selected tool and its instantiated arguments. This allows later calls to consume intermediate results (e.g., a timestamp returned by EVENTLOCATE can serve as a temporal constraint for a subsequent COMPUTE call).

Question Answering. Finally, the answer model generates response conditioned on the query and an evidence context comprising the instantiated plan and its tool outputs.

4 Experimental Evaluation

4.1 Experimental Setup

Benchmarks. We evaluate ADAMM on two complex multimodal memory benchmarks. **MemEye** evaluates visual-memory granularity and reasoning through paired multiple-choice and open-ended queries (Guo et al., 2026), while **MemGallery** assesses long-term conversational memory management (Bei et al., 2026). **MemEye** uses EM, BLEU-1 and LLM-judge for evaluation,

Benchmarks	Sessions	Rounds	Images	QA pairs
MemEye	221	848	438	742
MemGallery	240	3,962	1,003	1,711

Table 2: Benchmark statistics.

whereas **MemGallery** reports F1, BLEU-1, and LLM-Judge scores. Dataset statistics are provided in Table 2.

Baselines. We compare against unimodal memory agents, including **A-Mem** and **MemoryOS** (Xu et al., 2025; Li et al., 2025), which organize long-term textual interaction histories. Multimodal baselines include dedicated memory agents (**M2A**, **MMA**, and **MIRIX**) (Feng et al., 2026; Lu et al., 2026; Wang and Chen, 2025), which construct persistent memories from dialogue and visual observations, and retrieval-based systems (**MM-RAG** and **UniversalRAG**), which directly retrieve relevant multimodal evidence for answer generation.

Implementation details. We use GPT-4.1-nano and GPT-5.4-mini as answer and memory construction backbones for all methods. Text and image representations are produced by MiniLM-L6-v2 and siglip2-base-patch16-384, respectively. We use Top-10 retrieved memory units for retrieval memory baselines, and our method constrain the planner to generate at most three execution steps under a shared budget of 10 evidence units. Results are averaged over 3 runs.

4.2 Main Results

Overall performance. Table 3 presents a comprehensive comparison of ADAMM with representative unimodal and multimodal memory frameworks across two benchmarks and answer backbones. ADAMM consistently achieves the best result on every metric. With GPT-4.1-nano, it surpasses the strongest competing result on MemEye by 7.3% and 5.8% percentage points for MCQ and open-ended questions, respectively, and improves MemGallery F1, BLEU-1, and LLM-Judge by 4.2%, 4.3%, and 7.0%. The advantage persists with GPT-5.4-mini, yielding gains of 3.2% and 11.3% points on MemEye and 2.6%, 2.2%, and 5.2% on the three MemGallery metrics. The consistent improvements on different settings demonstrate that ADAMM generalizes across answer formats, benchmark settings, and backbone capacities.

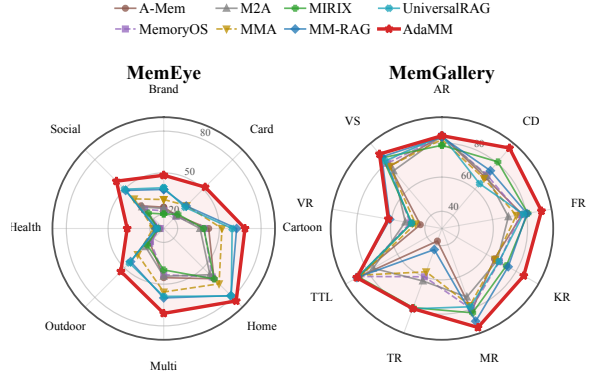


Figure 3: LLM-as-Judge performance across fine-grained tasks with GPT-5.4-mini.

Fine-grained task analysis. To further examine where the improvement arises, Figure 3 reports the LLM-Judge breakdown using GPT-5.4-mini. On MemEye, the largest gains over the strongest baseline occur on Card Playlog and Personal Health, with improvements of 18.8% and 16.7%, respectively. Both tasks require exact operations over recurring records, such as filtering observations, comparing values, or tracking changes, highlighting the benefit of analytic memory. ADAMM also achieves substantial gains on Multi-scene (11.1%), Brand Memory (9.0%), Outdoor Navigation (8.6%), and Social Chat (8.2%), where relevant evidence must be retrieved and connected across interactions. The smaller gains on Cartoon Entertainment (6.1%) and Home Renovation (5.2%) are consistent with their greater reliance on direct visual-semantic recall. On MemGallery, the largest margins appear in Conflict Detection (10.6%), Knowledge Resolution (10.5%), and Factual Retrieval (8.1%), demonstrating the value of structured access for reconciling and precisely locating information. ADAMM further improves Multi-entity Reasoning, Visual-centric Reasoning, and Visual-centric Search, while remaining comparable to the strongest baselines on Answer Refusal, Temporal Reasoning, and Test-time Learning. Together, these results suggest that combining retrieval with operation-specific analytic access is particularly beneficial when answering requires more than semantic relevance alone.

4.3 Ablation Study

We evaluate four ablated variants on MemEye to examine the contributions of the designed memories and the memory-aware planning mechanism. The Health task emphasizes numerical and tempo-

Backbone		Method	MemEye			MemGallery		
			EM	BLEU-1	LLM-Judge	F1	BLEU-1	LLM-Judge
GPT-4.1-nano	Unimodal	A-Mem	39.1	17.1	32.0	53.2	47.2	67.4
		MemoryOS	42.4	15.8	27.6	53.4	47.1	65.9
	Multimodal	M2A	34.1	5.8	13.8	50.2	44.3	61.5
		MMA	39.0	10.6	32.0	55.9	50.7	64.1
		MIRIX	40.9	5.7	13.6	56.3	51.1	66.3
		MM-RAG	42.7	16.6	42.2	58.3	52.4	67.3
		UniversalRAG	43.1	16.5	40.2	57.7	52.4	67.0
		ADAMM (Ours)	50.4(+7.3)	21.2(+4.1)	48.0(+5.8)	62.6(+4.2)	56.7(+4.3)	74.3(+7.0)
GPT-5.4-mini	Unimodal	A-Mem	48.0	22.4	35.2	63.0	58.6	74.4
		MemoryOS	48.7	21.8	33.3	64.5	59.0	75.5
	Multimodal	M2A	40.1	12.3	33.5	60.8	56.0	71.7
		MMA	53.9	25.7	43.3	64.0	59.6	73.3
		MIRIX	46.7	18.0	33.2	65.7	60.9	78.2
		MM-RAG	61.8	29.7	49.4	66.5	61.9	78.7
		UniversalRAG	62.4	30.4	48.6	65.6	60.8	77.4
		ADAMM (Ours)	65.5(+3.2)	35.9(+5.5)	60.7(+11.3)	69.1(+2.6)	64.1(+2.2)	83.9(+5.2)

Table 3: Main results on MemEye and MemGallery. Higher is better for all metrics. **bold values** denote the best result within a backbone and **red values** denote improvements over the strongest baseline.

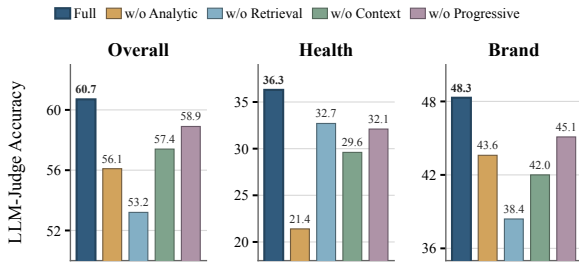


Figure 4: **Ablation results on MemEye.** Overall and task-level LLM-as-Judge accuracy using GPT-5.4-mini.

ral analysis over multiple personal-health records, whereas the Brand task primarily evaluates visually grounded recall of brand-related information. In *w/o Analytic* and *w/o Retrieval*, we disable the analytic memory and retrieval memory, respectively. In *w/o Planning Context*, the planner receives the entire memory metadata. In *w/o Progressive Execution*, all tool calls and their arguments are instantiated before execution.

Contributions of Complementary Memories.

As shown in Figure 4, removing either memory component consistently degrades performance. Without analytic memory, overall accuracy decreases by 4.6%, with a substantially larger 14.9% drop on Health. This result highlights the importance of organizing recurring records into executable structures for analytical queries. Removing retrieval memory instead causes a 7.5% overall drop and is particularly detrimental to Brand, where performance decreases by 9.9%. The distinct degra-

dation patterns confirm that analytic and retrieval memory provide complementary capabilities.

Contributions of Memory-Aware Planning.

Removing the planning context reduces performance by 3.3% overall and by 6.7% and 6.3% on Health and Brand, respectively. This demonstrates that effective tool selection requires knowledge of both the query intent and the memory structures currently available. Instantiating all calls before execution also produces consistent degradation, including drops of 4.2% on Health and 3.2% on Brand. These results indicate that progressive execution is useful when later operations depend on information obtained from preceding calls.

5 Conclusion

In this work, we identify a retrieval-analysis mismatch in long-term multimodal agent memory, highlighting the need for executable operations over accumulated experience beyond retrieval. To address this issue, we introduced ADAMM, which couples hierarchical retrieval memory with schema-induced analytic memory and exposes their distinct capabilities through operation-specific tools. A memory-aware planner further grounds tool selection in the current memory state and progressively composes retrieval and analytic operations. Experiments on MemEye and MemGallery across two answer backbones demonstrate consistent improvements from combining flexible recall with executable analysis.

Limitations

While ADAMM demonstrates promising results in long-term multimodal agent memory, two limitations remain. First, analytic memory depends on record fragments extracted from multimodal interactions. Incorrect or missing fields may propagate to schema induction, table construction, and downstream computations. Future work could incorporate confidence-aware extraction and cross-round consistency checks to improve robustness. Second, ADAMM uses a predefined set of access tools and thus requires manual extension for unseen, domain-specific operations. Future work could develop a self-evolving tool framework that identifies emerging capability gaps and safely synthesizes, validates, and integrates new tools.

References

- Rakesh Agrawal and Ramakrishnan Srikant. 1994. Fast algorithms for mining association rules in large databases. In *Proceedings of the 20th International Conference on Very Large Data Bases*, pages 487–499.
- Yuanchen Bei, Tianxin Wei, Xuying Ning, Yanjun Zhao, Zhining Liu, Xiao Lin, Yada Zhu, Hendrik Hamann, Jingrui He, and Hanghang Tong. 2026. Mem-gallery: Benchmarking multimodal long-term conversational memory for mllm agents. *arXiv preprint arXiv:2601.03515*.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. Mem0: Building production-ready ai agents with scalable long-term memory. *arXiv preprint arXiv:2504.19413*.
- Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, and 3 others. 2023. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378*.
- Junyu Feng, Binxiao Xu, Jiayi Chen, Mengyu Dai, Cenyang Wu, Haodong Li, Bohan Zeng, Yunliu Xie, Hao Liang, Ming Lu, and Wentao Zhang. 2026. M2a: Multimodal memory agent with dual-layer hybrid memory for long-term personalized interactions. *arXiv preprint arXiv:2602.07624*.
- Minghao Guo, Qingyue Jiao, Zeru Shi, Yihao Quan, Boxuan Zhang, Danrui Li, Liwei Che, Wujiang Xu, Shilong Liu, Zirui Liu, Mubbasir Kapadia, Vladimir Pavlovic, Jiang Liu, Mengdi Wang, Yiyu Shi, Dimitris N. Metaxas, and Ruixiang Tang. 2026. Memeye: A visual-centric evaluation framework for multimodal agent memory. *arXiv preprint arXiv:2605.15128*.
- Kostas Hatalis, Despina Christou, Joshua Myers, Steven Jones, Keith Lambert, Adam Amos-Binks, Zohreh Dannenhauer, and Dustin Dannenhauer. 2023. Memory matters: The need to improve long-term memory in llm-agents. In *Proceedings of the AAAI Symposium Series*, volume 2, pages 277–280.
- Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. 2024. Webvoyager: Building an end-to-end web agent with large multimodal models. *arXiv preprint arXiv:2401.13919*.
- Zhengjun Huang, Zhoujin Tian, Qintian Guo, Fangyuan Zhang, Yingli Zhou, Di Jiang, Zeying Xie, and Xiaofang Zhou. 2025. Licomemory: Lightweight and cognitive agentic memory for efficient long-term reasoning. *arXiv preprint arXiv:2511.01448*.
- Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. 2024. Visualwebarena: Evaluating multimodal agents on realistic visual web tasks. *arXiv preprint arXiv:2401.13649*.
- Yuanchun Li, Hao Wen, Weijun Wang, Xiangyu Li, Yizhen Yuan, Guohong Liu, Jiacheng Liu, Wenxing Xu, Xiang Wang, Yi Sun, et al. 2024. Personal llm agents: Insights and survey about the capability, efficiency and security. *arXiv preprint arXiv:2401.05459*.
- Zhiyu Li, Shichao Song, Hanyu Wang, Simin Niu, Ding Chen, Jiawei Yang, Chenyang Xi, Huayi Lai, Jihao Zhao, Yezhaohui Wang, Junpeng Ren, Zehao Lin, Jiahao Huo, Tianyi Chen, Kai Chen, Kehang Li, Zhiqiang Yin, Qingchen Yu, Bo Tang, and 3 others. 2025. Memos: An operating system for memory-augmented generation (MAG) in large language models. *CoRR*, abs/2505.22101.
- Jiaqi Liu, Zipeng Ling, Shi Qiu, Yanqing Liu, Siwei Han, Peng Xia, Haoqin Tu, Zeyu Zheng, Cihang Xie, Charles Fleming, Mingyu Ding, and Huaxiu Yao. 2026. Omni-simplemem: Autoresearch-guided discovery of lifelong multimodal agent memory. *arXiv preprint arXiv:2604.01007*.
- Yihao Lu, Wanru Cheng, Zeyu Zhang, and Hao Tang. 2026. Mma: Multimodal memory agent. *arXiv preprint arXiv:2602.16493*.
- Viet Nguyen, Thao Nguyen, Vishal M. Patel, and Yuheng Li. 2026. Personal visual memory from explicit and implicit evidence. *arXiv preprint arXiv:2605.28806*.
- Edward R. Omiecinski. 2003. [Alternative interest measures for mining associations in databases](#). *IEEE Transactions on Knowledge and Data Engineering*, 15(1):57–69.

- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. Memgpt: Towards llms as operating systems. *arXiv preprint arXiv:2310.08560*.
- Alex Petrov, Alexander Guskov, Denis Mukha, and Dima Korolev. 2026. [From unstructured recall to schema-grounded memory: Reliable AI memory via iterative, schema-aware extraction](#). *arXiv preprint arXiv:2604.27906*.
- Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A temporal knowledge graph architecture for agent memory, 2025. URL <https://arxiv.org/abs/2501.13956>.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *arXiv preprint arXiv:2302.04761*.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023. Voyager: An open-ended embodied agent with large language models. *arXiv preprint arXiv:2305.16291*.
- Xiyao Wang, Yuhang Zhou, Xiaoyu Liu, Hongjin Lu, Yuancheng Xu, Feihong He, Jaehong Yoon, Taixi Lu, Gedas Bertasius, Mohit Bansal, Huaxiu Yao, and Furong Huang. 2024. Mementos: A comprehensive benchmark for multimodal large language model reasoning over image sequences. *arXiv preprint arXiv:2401.10529*.
- Yu Wang and Xi Chen. 2025. Mirix: Multi-agent memory system for llm-based agents. *arXiv preprint arXiv:2507.07957*.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. 2024. Osvorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. *arXiv preprint arXiv:2404.07972*.
- Wujiang Xu, Kai Mei, Hang Gao, Juntao Tan, Zujie Liang, and Yongfeng Zhang. 2025. A-mem: Agentic memory for llm agents. *arXiv preprint arXiv:2502.12110*.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2022. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*.
- Juwei Yue, Chuanrui Hu, Jiawei Sheng, Zuyi Zhou, Wenyan Zhang, Tingwen Liu, Li Guo, and Yafeng Deng. 2026. Hypermem: Hypergraph memory for long-term conversations. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 35237–35254.
- Zeyu Zhang, Quanyu Dai, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2025. A survey on the memory mechanism of large language model-based agents. *ACM Transactions on Information Systems*, 43(6):1–47.
- Wanjuan Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. Memorybank: Enhancing large language models with long-term memory. In *AAAI*, pages 19724–19731. AAAI Press.